

DISPERSIVE POWER OF A PRISM – SPECTROMETER

AIM:

To determine the dispersive power of the material of a given prism by the spectrometer.

APPARATUS:

Spectrometer, crown glass prism and mercury vapour lamp.

FORMULA:

The dispersive power of the material of the given prism is expressed by the relation

$$\omega = \frac{\mu_2 - \mu_1}{\mu - 1}$$

Where μ_1 and μ_2 are the refractive indices of two colours.

And
$$\mu = \frac{\mu_1 + \mu_2}{2}$$

Usually the colours chosen are blue and red, so

$$\omega = \frac{\mu_b - \mu_r}{\mu - 1}$$

Where

$$\mu = \frac{\mu_b + \mu_r}{2}$$

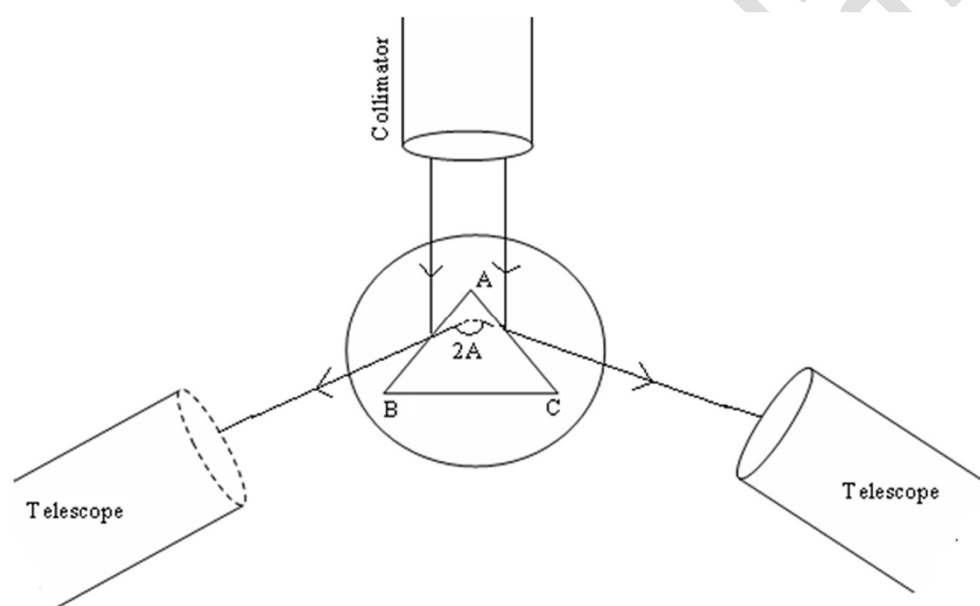
Preliminary Adjustments:

The primary adjustments of spectrometer can be done as given below

a) Eye – Piece: The telescope is turned towards a white surface i.e., wall and the Eye- piece is moved in (or) out until the cross-wires are seen clearly.

b) Telescope: The Telescope is directed towards a distant object that is telegraph pole (or) a tree. By adjusting the pinion, the telescope is adjusted until the image of the object is formed in the plane of cross-wires with no parallax error. Now the telescope is ready to receive a parallel beam of light.

c) Collimator: The slit of the collimator is illuminated with sodium light. The telescope is brought in line with the collimator and the distance of the slit from the collimating lens is adjusted until a clear image of the slit with well-defined edges is formed in the plane of the cross wires without any parallax error. Slit is adjusted to be vertical and narrow.



d) Prism table: A spirit level is kept on the prism table parallel to the line joining to the levelling screws. The two screws are adjusted until the air bubble of the spirit level comes to the centre. Then the spirit level is turned on the table perpendicular to this position and the third screw is adjusted until the air – bubble comes to the centre. Now the surface of the prism table will be horizontal.

$$\text{Least count } L.C. = \frac{1MS}{TOTAL VSD} = \frac{30'}{30} = 1'$$

PROCEDURE:

The prism is mounted on the prism table and the position of the prism is adjusted to observe the spectrum of the mercury vapour lamp. Observing the blue line in the spectrum through the telescope, the prism is adjusted for minimum deviation position. The tangent screw of the telescope is adjusted, the position of the prism is adjusted so that blue line is just on point of refracting its path after coming to the point of intersection of the cross wires. The readings of the telescope for the minimum deviation of red line are noted. The telescope is brought in line with the collimator and removing the prism, the direct readings on both verniers are noted. The respective difference gives the minimum deviation for blue and red colours and their refractive indices are found by

$$\mu_b = \frac{\sin\left[\frac{A+D_b}{2}\right]}{\sin\left[\frac{A}{2}\right]} \quad \text{and} \quad \mu_r = \frac{\sin\left[\frac{A+D_r}{2}\right]}{\sin\left[\frac{A}{2}\right]}$$

The dispersive power of the material of the prism for blue and red colours is found by the relation.

$$\omega = \frac{\mu_2 - \mu_1}{\mu - 1}$$

The readings are tabulated as shown.

OBSERVATIONS:

Angle of prism $A = 60^\circ$

Table:

S. No.	Colour of spectrum	Reading of Telescope		Direct Reading		Angle of Direction		Mean (D)	$\mu = \frac{\sin\left[\frac{A+D}{2}\right]}{\sin\left[\frac{A}{2}\right]}$
		V_1	V_2	V_1'	V_2'	$V_1 \sim V_1'$	$V_2 \sim V_2'$		
1.	Red								
2.	Blue								

CALCULATIONS:

PRECAUTIONS:

- (i) The Prism should be adjusted for each colour separately.
- (ii) Readings are noted without parallax error

RESULT:

Dispersive power of the material of the prism, $\omega =$ _____

Dr. S. Damodar